

7.3 Classwork: Logarithm and Graphing Challenge (no calculator)

1. **Change of base**

(a) Solve for x :

$$\log_2 x + \log_4 x + \log_8 x = 11.$$

(b) Show that, for $a, b > 0$ with $a, b \neq 1$,

$$\log_a b \cdot \log_b a = 1.$$

(c) Given that $\log_a b = 4$, find the exact value of $\log_{a^2} b^3$.

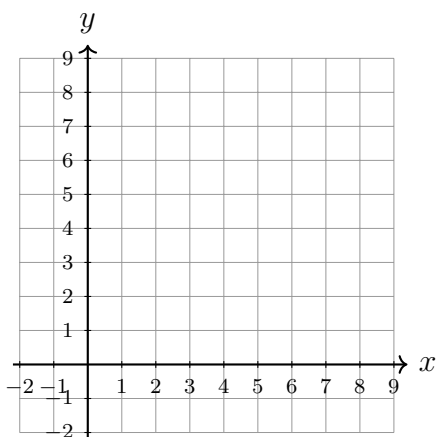
2. **Quadratic in log.** Solve for x :

$$(\log_3 x)^2 - 5\log_3 x + 6 = 0.$$

3. **An exponential and its inverse.** Let $f(x) = 3^{x-1} + 2$.

(a) Find $f^{-1}(x)$.

(b) On the axes below, sketch $y = f(x)$, $y = f^{-1}(x)$, and $y = x$. Label each curve and the asymptote of each.



4. **Solve for the exponent.** When x is in the exponent, take the logarithm of both sides. Give exact answers.

(a) Solve $2^x = 20$.

(b) Solve $5^{x-1} = 12$.

(c) A bacteria culture is modelled by $P(t) = 50 \cdot 2^t$, where t is the time in hours. Find the exact value of t at which $P = 1500$.

(d) Solve $5^{2x} = 3^{x+1}$.